Research Collaborator Recommendation

Social Network Analysis AAT | Executed experiment on Stanford SNAP ca-GrQc

Abstract

This project compares four structural link prediction scores on a real research collaboration network. The cleaned graph has 5,242 researchers and 14,484 observed co-authorship links. A reproducible random edge split evaluates missing-link recovery. Adamic-Adar has the highest primary ROC-AUC in this experiment. Separate exact rankings on the full graph suggest potential collaborator pairs. A notebook and local Streamlit app expose the same calculations.

Problem statement and objectives

Given only the observed co-authorship structure, rank unconnected researcher pairs. Objectives: construct a clean graph, compare Common Neighbors, Jaccard, Adamic-Adar and Preferential Attachment, evaluate reproducibly, and explain top 10 and top 20 structural recommendations. Scores do not measure topic compatibility.

Dataset and preparation

Source: Stanford SNAP ca-GrQc. The archive covers arXiv GR-QC papers from January 1993 to April 2003. Nodes are integer researcher IDs. Edges represent co-authorship, with no weights or directions. The preserved original contains 28,980 rows. Cleaning removes 12 self-loop rows and 14,484 duplicate/reverse rows. All nodes remain available. SHA256 is recorded in the dataset README.

SNAP lists 14,496 edges. The downloaded file contains 12 self-loop rows, so the required simple loop-free graph has 14,484 edges after cleaning. The report consistently uses the cleaned count.

Statistic	Full cleaned graph
nodes	5242
edges	14484
components	355
largest component	4158
isolates	1
density	0.001054
degree min	0
degree max	81
degree mean	5.526135
degree median	3.000000
average clustering	0.529636

Methodology and formulas

Common Neighbors: Counts shared collaborators. CN(u,v) = |N(u) intersection N(v)|

Jaccard: Divides shared collaborators by the size of the combined collaborator set. J(u,v) = |N(u) intersection N(v)| / |N(u) union N(v)|

Adamic-Adar: Gives more weight to shared collaborators who have fewer connections. AA(u,v) = sum over shared w of 1 / ln(d(w))

Preferential Attachment: Multiplies the two researchers' degrees, favoring well-connected researchers. PA(u,v) = d(u) * d(v)

N(u) is the neighbor set and d(u) its size. The Jaccard score is zero for an empty union. A shared neighbor of distinct nodes has degree at least two, so Adamic-Adar is defined. Natural logarithms are used. Custom scores are checked against NetworkX on every evaluation pair, with Common Neighbors checked against nx.common_neighbors.

Experimental setup

Primary seed: 42. Sort unique unordered edges, shuffle with random.Random, take floor(80%) for training, and hold out the rest. Add all original nodes to the training graph before edges. Negatives are sampled without replacement from pairs absent from the complete graph, with no self-pairs. Use one negative per held-out positive. Scoring uses only the training graph. The full graph is consulted solely to reject false negatives. Every method sees the same evaluation pairs. No parameters are tuned.

Training has 11,587 edges and 5,242 nodes. Holding out edges changes components from 355 to 650, largest component size from 4158 to 3875, and isolates from 1 to 298. Training disconnection is allowed, not repaired.

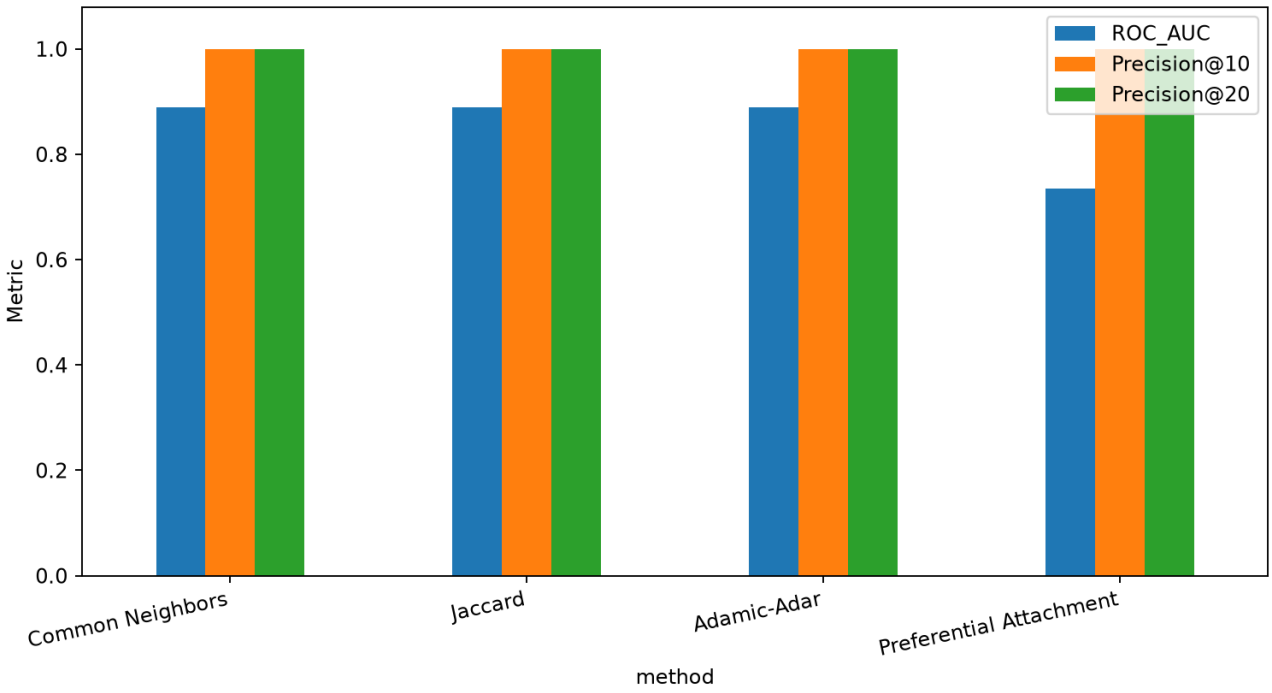
Deterministic ranking sorts by descending score then ascending smaller ID and larger ID. ROC-AUC compares positive/negative ordering and gives tied scores half credit. Precision@K = hits/K. Recall@K = hits/number of held-out positives. F1@K = 2PR/(P+R), with zero if both are zero. Recall and F1 are explicitly at K. Balanced sampled-set precision is not precision over every missing link.

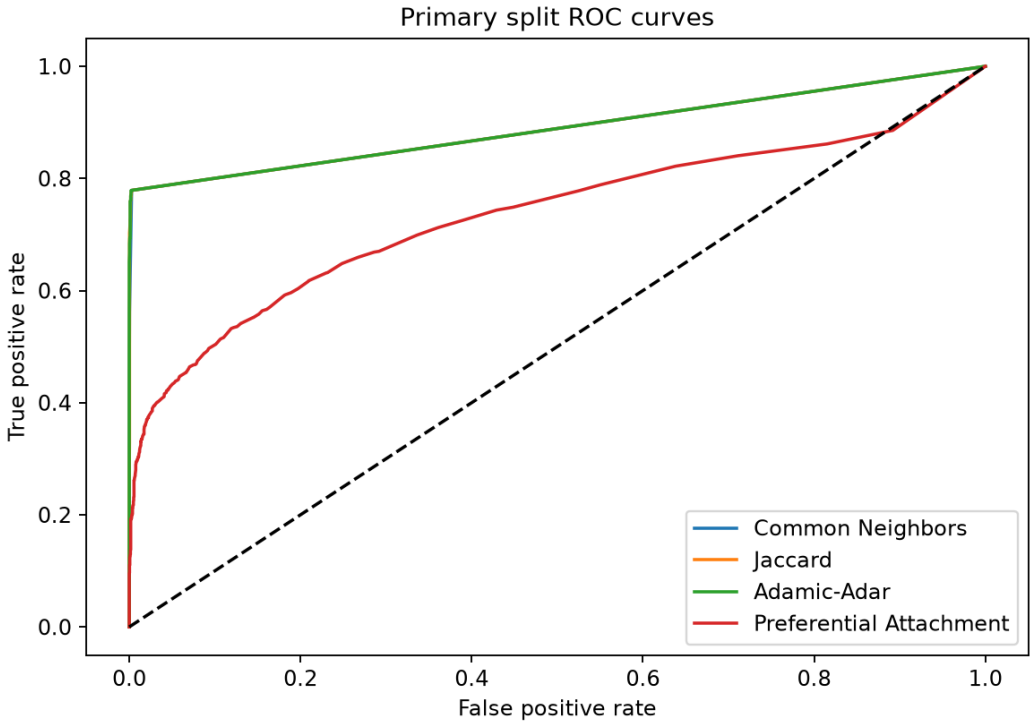
Actual primary results

Method	ROC_AUC	Precision@10	Precision@20
Common Neighbors	0.88870	1.00000	1.00000
Jaccard	0.88882	1.00000	1.00000
Adamic-Adar	0.88894	1.00000	1.00000
Preferential Attachment	0.73502	1.00000	1.00000

Method	Recall@10	Recall@20	F1@10	F1@20
Common Neighbors	0.00345	0.00690	0.00688	0.01371
Jaccard	0.00345	0.00690	0.00688	0.01371
Adamic-Adar	0.00345	0.00690	0.00688	0.01371
Preferential Attachment	0.00345	0.00690	0.00688	0.01371

Selection criterion: highest primary ROC-AUC, with alphabetical method name breaking any tie. The strongest method by this criterion is Adamic-Adar. The three neighborhood methods have nearly equal AUCs on this split. All four methods achieve Precision@10 and Precision@20 of 1.0 on the sampled set. This is a descriptive choice on the same test results, not independent validation. High Precision@K can coexist with low Recall@K because only 10 or 20 predictions cover thousands of positives. Preferential Attachment favors popular researchers; the neighborhood scores reward shared structure.





Split sensitivity

Method	Mean AUC	SD	Min	Max
Adamic-Adar	0.89772	0.00611	0.88894	0.90272
Common Neighbors	0.89747	0.00612	0.88870	0.90259
Jaccard	0.89756	0.00607	0.88882	0.90255
Preferential Attachment	0.73897	0.00468	0.73495	0.74402

Summary includes seeds 42, 7, 21 and 84. Each seed changes both the split and sampled negative set. Four runs provide a small sensitivity check, not a confidence interval or independent final test.

Final collaborator recommendations

After evaluation, the shared pipeline scores all 13,722,177 unordered nonedges of the full cleaned graph. It streams all pairs with four heaps of 20 records, avoiding a large all-pairs table. There is no candidate filtering or approximation. Existing links, self-links and duplicate unordered pairs are excluded. The top-10 CSV is the first ten rows per method of the top-20 CSV.

Common Neighbors

Rank	ID 1	ID 2	Score	Shared
1	45	46	42.00000	42
2	4511	4513	42.00000	42
3	4511	22691	42.00000	42
4	46	4513	41.00000	41
5	46	22691	41.00000	41
6	5210	7444	25.00000	25
7	7444	8727	25.00000	25
8	7444	9869	25.00000	25
9	7444	12710	25.00000	25
10	7444	15245	25.00000	25
11	7444	15933	25.00000	25
12	7444	19446	25.00000	25
13	7444	20576	25.00000	25
14	7444	22504	25.00000	25
15	2449	5210	24.00000	24
16	2449	8727	24.00000	24
17	2449	12710	24.00000	24
18	2449	15245	24.00000	24
19	2449	15457	24.00000	24
20	2449	19446	24.00000	24

Jaccard

Rank	ID 1	ID 2	Score	Shared
1	29	16751	1.00000	1
2	85	5413	1.00000	1
3	114	2348	1.00000	1
4	114	6631	1.00000	1
5	123	23240	1.00000	1
6	195	19168	1.00000	3
7	232	1075	1.00000	1
8	232	13481	1.00000	1
9	232	16742	1.00000	1
10	252	15418	1.00000	1
11	312	16545	1.00000	2
12	352	20533	1.00000	3
13	374	20776	1.00000	1
14	392	3173	1.00000	1
15	392	3853	1.00000	1
16	392	15251	1.00000	1
17	392	16261	1.00000	1
18	392	21866	1.00000	1
19	400	14498	1.00000	1
20	628	13641	1.00000	1

Adamic-Adar

Rank	ID 1	ID 2	Score	Shared
1	45	46	10.59780	42
2	4511	4513	10.59205	42
3	4511	22691	10.59205	42
4	46	4513	10.33511	41
5	46	22691	10.33511	41
6	5210	7444	6.69908	25
7	7444	8727	6.69908	25
8	7444	9869	6.69908	25
9	7444	12710	6.69908	25
10	7444	15245	6.69908	25
11	7444	15933	6.69908	25
12	7444	19446	6.69908	25
13	7444	20576	6.69908	25
14	7444	22504	6.69908	25
15	2449	5210	6.39175	24
16	2449	8727	6.39175	24
17	2449	12710	6.39175	24
18	2449	15245	6.39175	24
19	2449	15457	6.39175	24
20	2449	19446	6.39175	24

Preferential Attachment

Rank	ID 1	ID 2	Score	Shared
1	15244	21012	4779.00000	0
2	15244	21281	4661.00000	0
3	12365	15244	4543.00000	0
4	15244	22691	4543.00000	0
5	6610	15244	4012.00000	0
6	9785	15244	4012.00000	0
7	6512	21012	3969.00000	0
8	15244	21508	3953.00000	0
9	15244	17655	3894.00000	0
10	6512	21281	3871.00000	0
11	2741	15244	3835.00000	0
12	6512	12365	3773.00000	0
13	6512	22691	3773.00000	0
14	15244	19423	3717.00000	0
15	15003	15244	3658.00000	0
16	13801	21012	3645.00000	1
17	13929	21012	3645.00000	0
18	13801	21281	3555.00000	1
19	13929	21281	3555.00000	1
20	14807	15244	3540.00000	0

Structural explanations

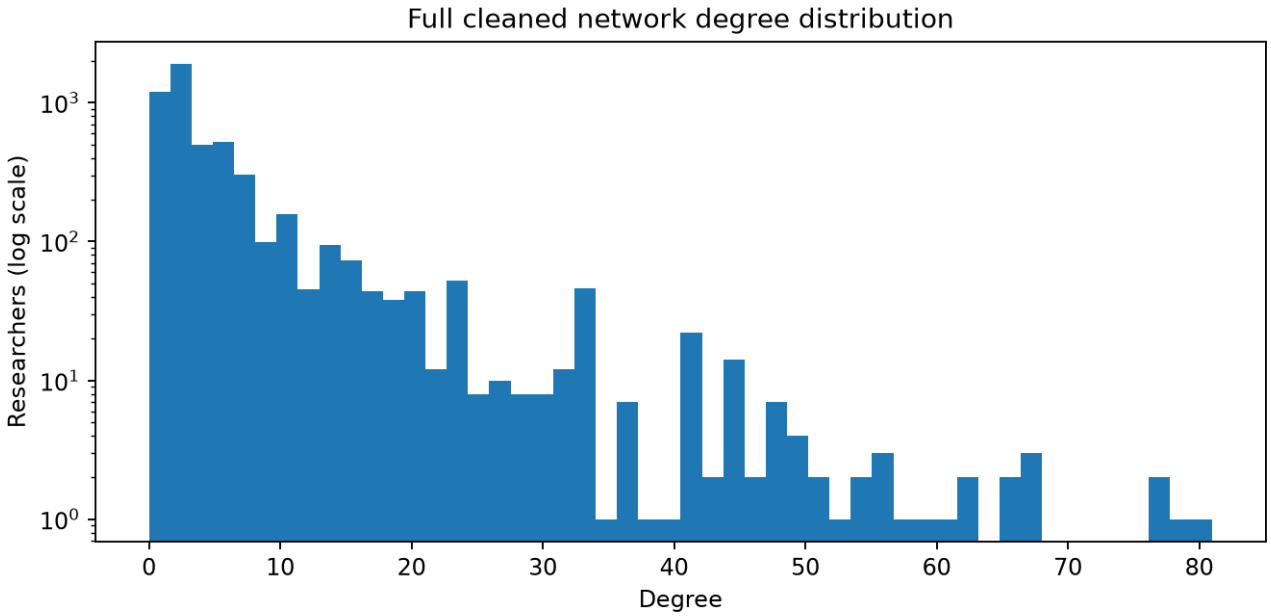
IDs 45 and 46: score 10.5978; degrees 49 and 42; 42 shared collaborators. Actual shared IDs: 570, 773, 1653, 2212, 2741, 2952, 3372, 4164, 4511, 6179, 6610, 6830, 7956, 8879, 9785, 11241, 11472, 12365, 12496, 12781, 12851, 14540, 14807, 15003, 15659, 17655, 17692, 18894, 19423, 19961, 20108, 20562, 20635, 21012, 21281, 21508, 21847, 22887, 23293, 24955, 25346, 25758. These nodes have no observed co-authorship edge. The score describes graph structure only.

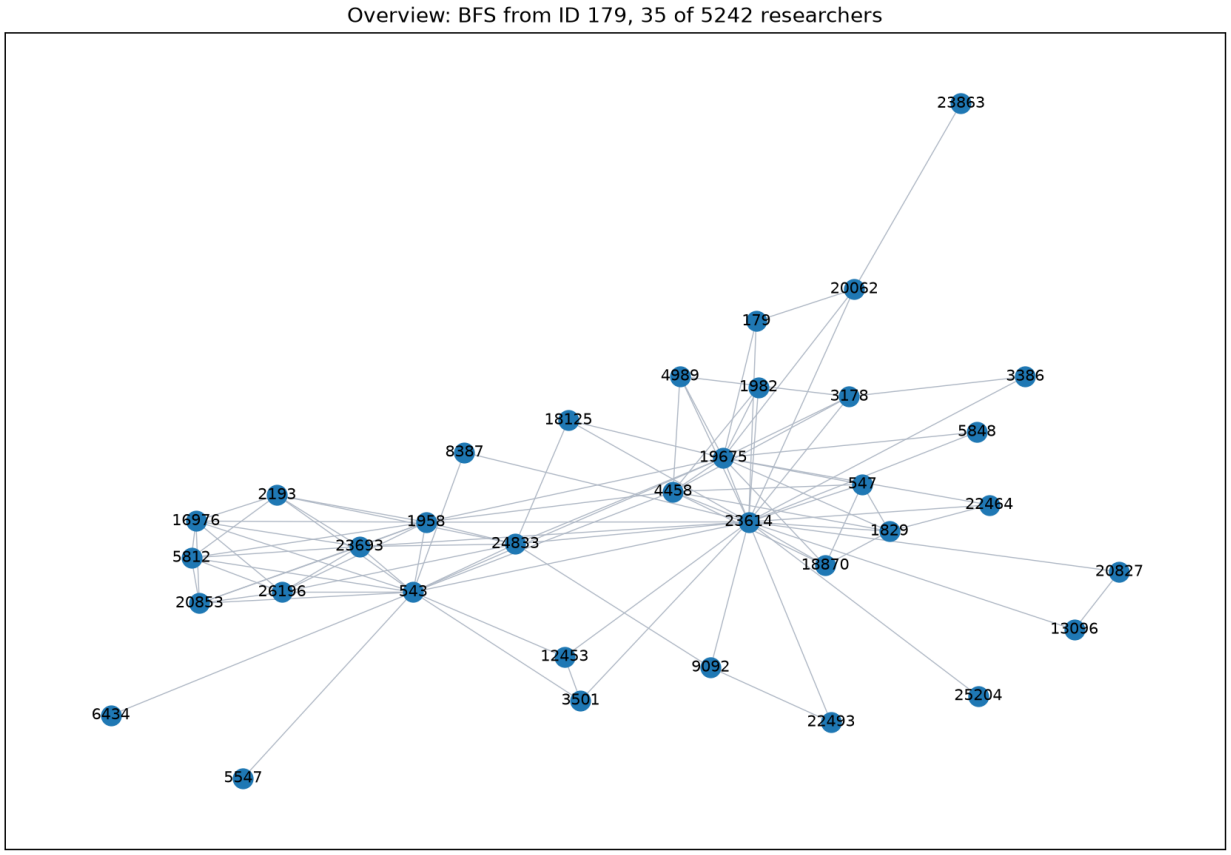
IDs 4511 and 4513: score 10.5921; degrees 45 and 43; 42 shared collaborators. Actual shared IDs: 45, 570, 773, 1653, 2212, 2741, 2952, 3372, 4164, 6179, 6610, 6830, 7956, 8879, 9785, 11241, 11472, 12365, 12496, 12781, 12851, 14540, 14807, 15003, 15659, 17655, 17692, 18894, 19423, 19961, 20108, 20562, 20635, 21012, 21281, 21508, 21847, 22887, 23293, 24955, 25346, 25758. These nodes have no observed co-authorship edge. The score describes graph structure only.

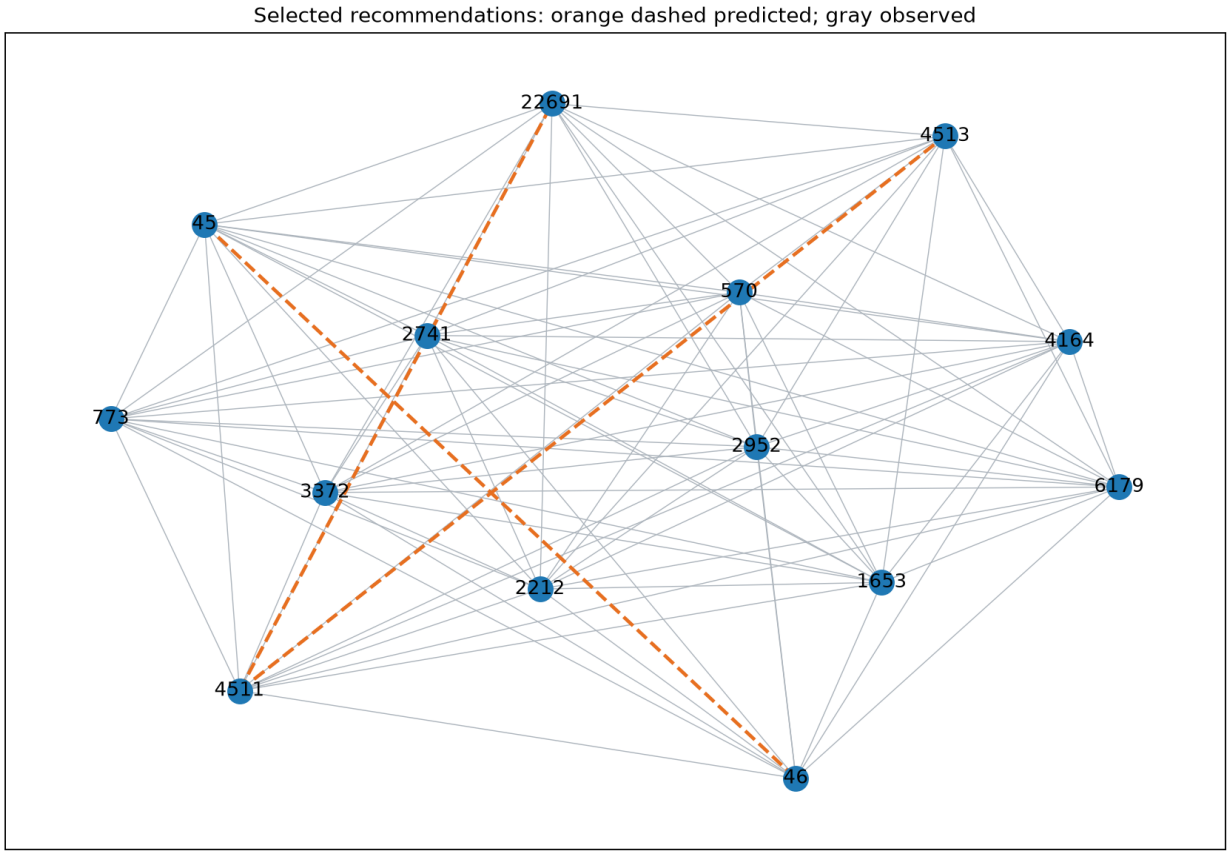
IDs 4511 and 22691: score 10.5921; degrees 45 and 77; 42 shared collaborators. Actual shared IDs: 45, 570, 773, 1653, 2212, 2741, 2952, 3372, 4164, 6179, 6610, 6830, 7956, 8879, 9785, 11241, 11472, 12365, 12496, 12781, 12851, 14540, 14807, 15003, 15659, 17655, 17692, 18894, 19423, 19961, 20108, 20562, 20635, 21012, 21281, 21508, 21847, 22887, 23293, 24955, 25346, 25758. These nodes have no observed co-authorship edge. The score describes graph structure only.

Visualizations

The degree distribution uses all nodes. The overview is a deterministic BFS subset of at most 35 researchers rooted at the lowest researcher ID with degree 3 in the largest component, with degree then ID ordering. The recommendation visualization contains three top pairs for the primary-AUC winner and up to ten shared collaborators per pair. Gray edges are observed and orange dashed edges are predicted. Full analysis always uses the entire graph.







network.html embeds its JavaScript and supports offline interaction. GraphML and CSV exports support Cytoscape. Cytoscape was unavailable on PATH, so no Cytoscape render was produced. See dataset/README.md for import and edge styling instructions.

Streamlit application

The local app provides Overview, Network Explorer, Algorithm Comparison, Collaborator Recommendations, Researcher Profile, and About and Limitations. Graph loading and per-researcher scores are cached. Global rankings come from precomputed CSVs. Researcher IDs come from validated selectors. Neighborhood views are bounded and clearly labeled. CSV downloads expose global and personal recommendations. Launch with python -m streamlit run app.py from the project folder.

Limitations

IDs have no supplied names, affiliations or interests. Recommendations are structural possibilities, not confirmed collaborations or assessments of research compatibility. Sampled nonedges are assumed negatives. A random edge split tests recovery of missing observed links, not future collaboration prediction. Balanced sampled precision differs from precision across all missing links. The best method is selected on these same test results, without independent validation.

The data is historical and may contain incomplete author coverage or identity ambiguity. Uniform random negatives can be easier than plausible nearby nonedges. Edge splitting can remove bridges and isolate nodes. The experiment has no temporal validation, topic features, or human assessment. Similarity scores are not calibrated probabilities.

Conclusion

The project supplies a reproducible comparison and exact structural recommendation rankings. Adamic-Adar leads on primary ROC-AUC under the stated protocol. A future extension could use timestamped papers and harder negative sets, with a separate validation/test design and topic evidence.

References

Stanford SNAP. General Relativity and Quantum Cosmology collaboration network. https://snap.stanford.edu/data/ca-GrQc.html

J. Leskovec, J. Kleinberg and C. Faloutsos (2007). Graph Evolution: Densification and Shrinking Diameters. ACM TKDD 1(1).

D. Liben-Nowell and J. Kleinberg (2007). The link-prediction problem for social networks. JASIST 58(7), 1019-1031. https://www.cs.cornell.edu/home/kleinber/link-pred.pdf

NetworkX documentation: Jaccard coefficient, Adamic-Adar index, Preferential Attachment, and common_neighbors. https://networkx.org/documentation/stable/reference/algorithms/link_prediction.html